

The value of this call option at time 0 is \$0.0428.²⁷

There is also a European put option with strike \$0.4660 maturing at time 2. This put option is worth \$0.0643 at time 0 and has the following payoffs at time 2:

$$\begin{aligned} P_{uu} &= \max(0.4660 - 4.0000, 0) = 0, \\ \text{or } P_{ud} &= \max(0.4660 - 4.0000, 0) = 0 = P_{du}, \\ \text{or } P_{dd} &= \max(0.4660 - 0.2500, 0) = 0.2160. \end{aligned} \tag{4.22}$$

Now compare the following strategies:

	time 0	time 2		
Strategy		$S_{uu} = 4$	$S_{ud} = S_{du} = 1$	$S_{dd} = 0.5$
Sell Put	+0.0643	0	0	-0.2160
Sell Call	+0.0428	-0.3240	0	0
Buy Bonds	-0.1071	0.1296	0.1296	0.1296
Buy-and-Hold Strategy	1.0000	2.8840	1.0840	0.6340
Short Volatility + Bonds	1.0000	2.6896	1.2136	0.5476
+ Buy and Hold				
Rebalanced Strategy	1.0000	2.6896	1.2136	0.5476

The table lists the values today in the column labeled “time 0” and the payouts of the various strategies at “time 2.” The time 2 payouts are contingent on the stock values at time 2; hence, there are three columns representing the stock values $S_{uu} = 4$, $S_{ud} = S_{dd} = 1$, and $S_{dd} = 0.5$ at time 2.

Consider the first set of strategies. Selling a put today means money comes in (+ sign) with a put premium of \$0.0643. If the stock price is low ($S_{dd} = 0.5$) at time 2, then the investor must pay out (– sign) an amount of \$0.2160. Likewise, selling a call today means money comes in with a call premium of \$0.0428. The investor must make a payout to the person buying the option of \$0.3240 if the stock increases at time 2 ($S_{uu} = 4$). The investor also purchases \$0.1071 of bonds at time 0. The purchase means a cash outflow, so there is a negative sign. At time 2, these bonds are worth $0.1071 \times (1.1)^2 = 0.1296$ at time 2. Finally, we have the payoffs of the buy-and-hold strategy starting with \$1 invested at time 0. The payoffs of the strategy in each state of the world for the buy-and-hold strategy are listed in equation (4.19).

If we add the short call, the short put, the long bond position, and the buy-and-hold strategy, we get a value of \$1 today at time 0, with identical payoffs to

²⁷This can be valued using *risk-neutral pricing*. The call value at time 0 is $\frac{q^2 \times 0.324}{(1.1)^2} = 0.0428$, where q is the *risk-neutral probability* given by $q = \frac{1.1-0.5}{2-0.5} = 0.4$. For an introduction to risk-neutral option pricing, see Bodie, Kane, and Marcus (2011). The put value in equation (4.22) is worth $\frac{(1-q)^2 \times 0.216}{(1.1)^2} = 0.0643$ at time 0.